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Semester and Department **2nd EE**

Lab report **3**

1. Aim

The goal of this laboratory exercise is to learn and apply the following concepts:

- Comparison (Relational) Operators
- Logical Operators
- Conditional Statements (if, if-else, else-if)

2. Background Theory

2.1 Comparison (Relational) Operators

Relational operators are used to compare two values. The result of the comparison is either true or false. These operators are commonly used in decision-making.

Common relational operators include:

- < : Smaller than
 - > : Greater than
 - <= : Smaller than or equal to
 - >= : Greater than or equal to
 - == : Equal to
 - != : Not equal to
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2.2 Logical Operators

Logical operators are used to connect multiple conditions or reverse a condition.

Main logical operators:

- ! : Logical NOT
- && : Logical AND
- || : Logical OR

These operators help in forming complex decision expressions.

2.3 Conditional Statements

Conditional statements control the flow of a program depending on certain conditions.

- **if statement:** Executes a block of code only when the condition is true.
 - **if-else statement:** Executes one block if the condition is true; otherwise, it executes another block.
 - **else-if ladder:** Used when there are multiple conditions to check in sequence.
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3. Programs and Outputs

Task 1: Using Comparison Operators

```
#include <iostream>
using namespace std;

int main()
{
    int a, b;

    cout << "Enter two values: ";
    cin >> a >> b;
```

```
    if (a == b)
        cout << "Both values are the same." << endl;
    else if (a > b)
        cout << "The first value is larger." << endl;
    else
        cout << "The first value is smaller." << endl;

    return 0;
}
```

Sample Output:

```
Enter two values: 10 20
The first value is smaller.
```

Task 2: Using Logical Operators

```
#include <iostream>
using namespace std;

int main()
{
    int number;

    cout << "Enter a number: ";
    cin >> number;

    if (number > 0 && number < 100)
        cout << "The number lies between 0 and 100." << endl;
    else
        cout << "The number does not satisfy the condition." << endl;

    return 0;
}
```

Sample Output:

```
Enter a number: 50
The number lies between 0 and 100.
```

Task 3: Simple if Statement

```
#include <iostream>
using namespace std;

int main()
{
    int value;

    cout << "Enter a number: ";
```

```
    cin >> value;

    if (value > 0)
        cout << "The number is positive." << endl;

    return 0;
}
```

Sample Output:

Enter a number: -5
(No output because the condition is false)

Task 4: if-else Statement

```
#include <iostream>
using namespace std;

int main()
{
    int age;

    cout << "Enter your age: ";
    cin >> age;

    if (age >= 18)
        cout << "You are allowed to vote." << endl;
    else
        cout << "You are not allowed to vote." << endl;

    return 0;
}
```

Sample Output:

Enter your age: 16
You are not allowed to vote.

Task 5: else-if Statement

```
#include <iostream>
using namespace std;

int main()
{
    int value;

    cout << "Enter a number: ";
    cin >> value;

    if (value > 0)
```

```
        cout << "The number is positive." << endl;
    else if (value < 0)
        cout << "The number is negative." << endl;
    else
        cout << "The number is zero." << endl;

    return 0;
}
```

Sample Output:

```
Enter a number: 0
The number is zero.
```

4. Explanation

- Comparison operators are essential for checking relationships between values.
- Logical operators allow combining multiple conditions in a single expression.
- Decision-making statements help programs choose different paths based on user input.

These concepts are widely used in programming to create interactive and intelligent applications.

5. Final Summary

In this laboratory session, I practiced implementing comparison operators, logical operators, and various conditional structures in C++. These elements are fundamental for controlling program flow and making decisions based on given conditions.